CASE STUDY DOCUMENT

Job Portal with Recruiter and Candidate Dashboard

HireConnect

Bridging Talent with Opportunity

Version 1.0 | 2026

1. Synopsis

HireConnect is a full-featured online job portal designed to bridge the gap between recruiters and job seekers. The platform supports two primary roles — Recruiters (employers) and Candidates (job seekers) — each with a dedicated, role-specific dashboard. The system streamlines the entire hiring lifecycle, from job posting and application submission to interview scheduling and offer management.

Recruiters can post job openings, review applications, shortlist candidates, schedule interviews, and manage the hiring pipeline. Candidates can create detailed profiles, search and filter job listings, submit applications, track their application status, and manage interview schedules — all from a single, unified interface.

HireConnect also incorporates notification services, resume parsing, analytics dashboards, and role-based access control to ensure a seamless and efficient hiring experience for all stakeholders.

2. Detailed Requirements

2.1 General Requirements

- Users can browse and search for job listings without logging in.
- Users must register and log in (via email/password or GitHub OAuth) to apply for jobs or post listings.
- Role-based access control: Recruiter role and Candidate role with distinct permissions.
- Both user types can update their profiles at any time.
- Users receive in-app and email notifications for key events.
- Admin panel for platform management and analytics.

2.2 Candidate Requirements

- Candidates can register, log in, and create a detailed professional profile.
- Upload and manage resumes (PDF format); system auto-parses key details.
- Search and filter jobs by title, location, salary range, experience level, and category.
- Apply to jobs with one click using saved profile details or a custom cover letter.
- Track application status in real time: Applied, Shortlisted, Interview Scheduled, Offered, Rejected.
- Receive interview invitations and confirm or reschedule from the dashboard.
- View and manage saved/bookmarked job listings.
- Access history of all previous applications and their outcomes.

2.3 Recruiter Requirements

- Recruiters can register, log in, and set up a company profile.
- Post new job openings with detailed descriptions, required skills, salary range, and job type.
- Manage posted jobs: edit, pause, close, or delete listings.

- View and filter incoming applications for each job posting.
- Shortlist, reject, or advance candidates through the hiring pipeline.
- Schedule interviews and send invitations directly to candidates.
- Download candidate resumes from the dashboard.
- Access analytics: number of applications, view-to-apply ratio, time-to-hire metrics.
- Manage team members who can co-manage job postings.

2.4 Notification and Communication

- Email notifications triggered on application status change, interview scheduling, and new job alerts.
- In-app notification centre for real-time updates.
- Recruiters can send messages to shortlisted candidates through the portal.

2.5 Payment and Subscription

- Freemium model: Candidates use the platform for free; Recruiters subscribe for premium features.
- Recruiter subscription tiers: Free (limited posts), Professional, and Enterprise.
- Payments via wallet, debit/credit card, or UPI.
- Recruiter billing dashboard with invoice history.

3. Use Case Diagram

The diagram below captures all actors and use cases within the HireConnect platform, spanning authentication, job management, application tracking, interview scheduling, notifications, and administrative controls.

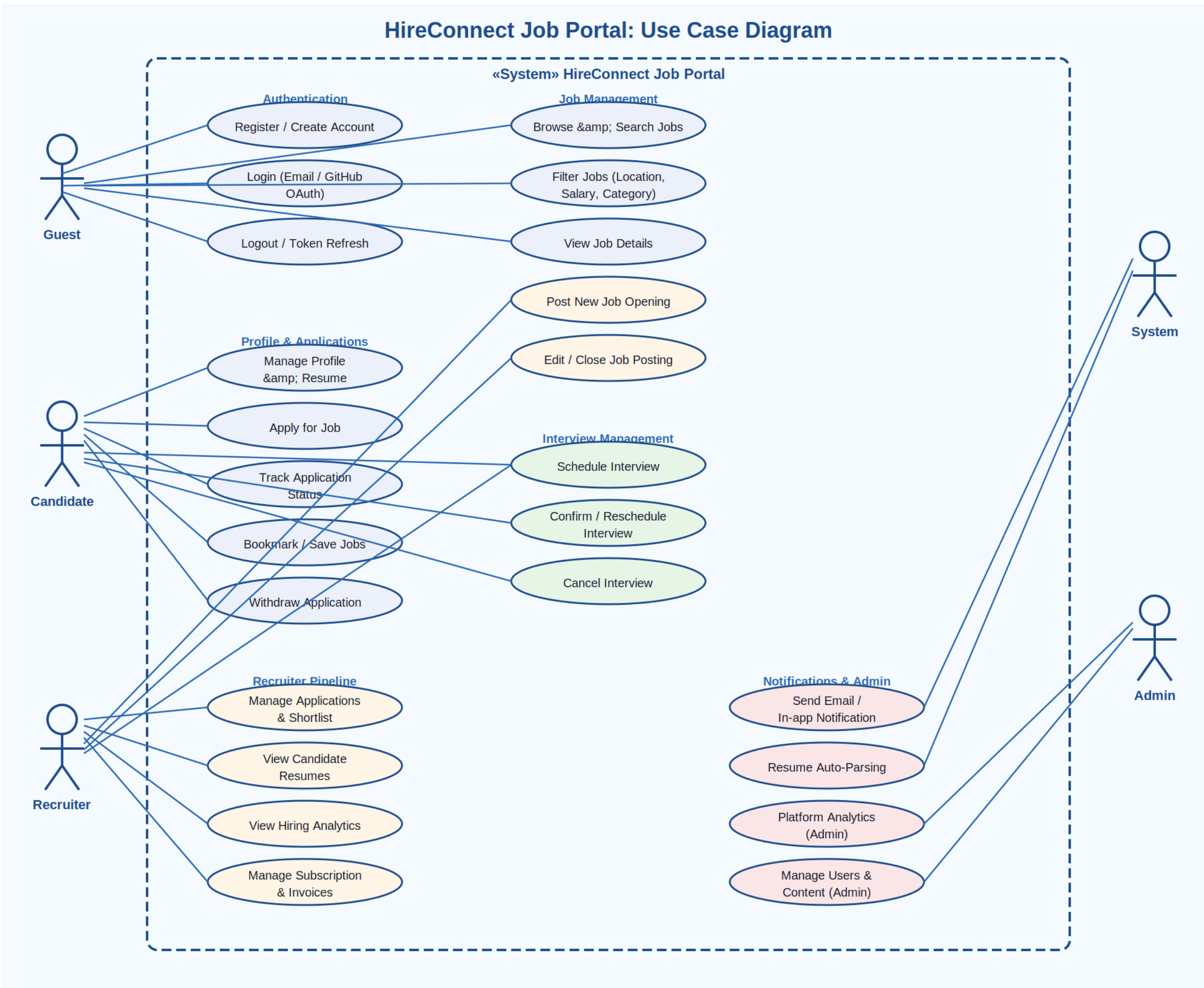

Figure 1: HireConnect Job Portal — Complete Use Case Diagram

3.1 Actors Summary

Actor	Description
Guest User	Unauthenticated visitor who can browse and search job listings
Candidate	Registered job seeker who applies for jobs and manages their profile
Recruiter	Employer who posts jobs and manages the entire hiring pipeline
System	Automated system handling notifications, parsing, and scheduling
Admin	Platform administrator with full access to manage users and content

3.2 Use Case Summary

Use Case	Actor(s)	Description
Register / Login	Candidate, Recruiter	Create account or log in via email or GitHub OAuth
Search Jobs	Guest, Candidate	Search and filter job listings by various criteria
Apply for Job	Candidate	Submit application using profile resume and cover letter
Track Application	Candidate	View real-time status of all submitted applications
Schedule Interview	Candidate, Recruiter	Confirm, reschedule, or cancel interview invitations
Post Job	Recruiter	Create and publish a new job opening on the portal
Manage Applications	Recruiter	Review, filter, shortlist, or reject incoming applications
Send Notification	System	Trigger emails and in-app alerts for status changes
View Analytics	Recruiter, Admin	Access hiring metrics and platform usage reports
Manage Subscription	Recruiter	Upgrade, downgrade, or cancel recruiter subscription plan

4. Class Diagrams — All Services

HireConnect is built as a microservices architecture. Each service follows the layered pattern: POJO/Entity → Repository Interface → Service Interface → ServiceImpl → REST Resource. The nine class diagrams below detail each service's domain model and component relationships.

4.1 Auth-Service

The Auth-Service manages all authentication concerns: user registration, login, logout, JWT token generation and validation, and OAuth2 integration with GitHub. It is the entry point for all secured operations.

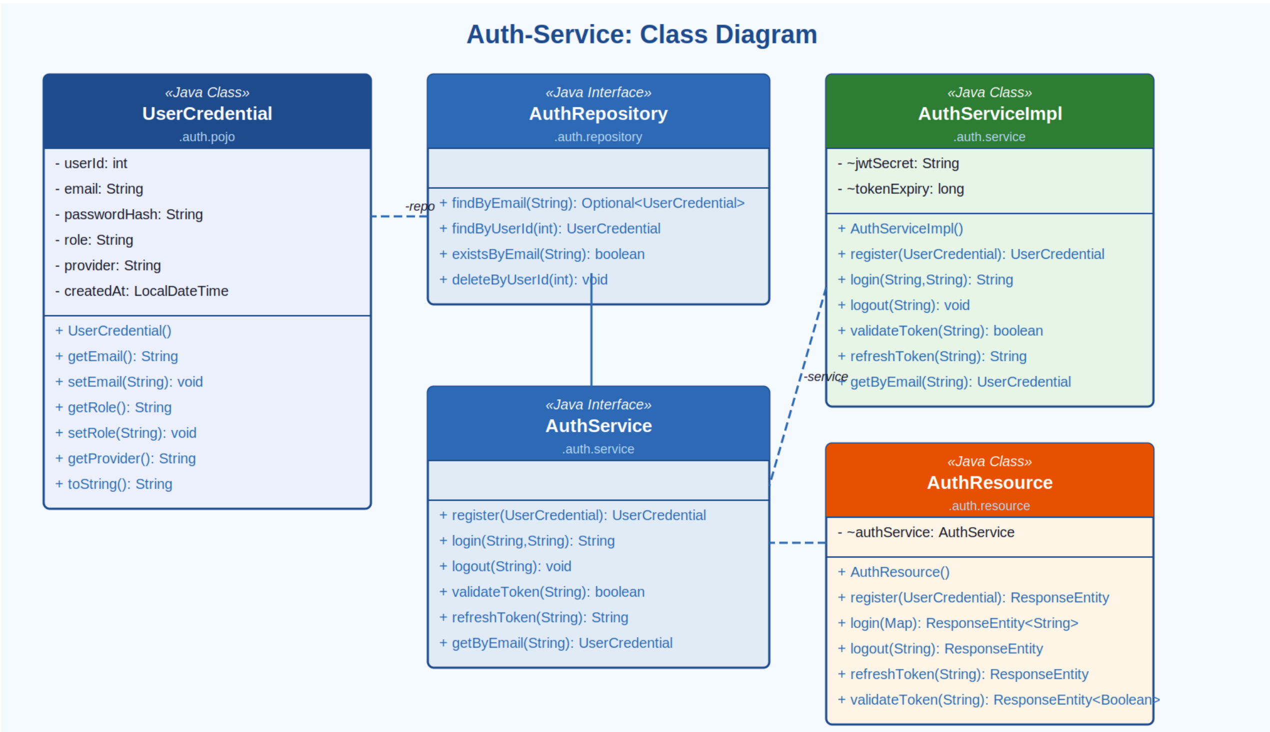

Figure 2: Auth-Service — Class Diagram

Component	Type	Responsibility
UserCredential	Java Class (POJO)	Stores userId, email, passwordHash, role, provider, createdAt
AuthRepository	Java Interface	findByEmail(), findByUserId(), existsByEmail(), deleteByUserId()
AuthServiceImpl	Java Class	Implements login, registration, JWT generation, OAuth handling
AuthService	Java Interface	Declares register(), login(), logout(), validateToken(), refreshToken()
AuthResource	Java Class (REST)	Exposes /auth/register, /auth/login, /auth/logout, /auth/refresh

4.2 Profile-Service

The Profile-Service manages candidate and recruiter profiles. It stores personal details, professional information, resume URLs, and delivery addresses. Both CandidateProfile and RecruiterProfile extend from a common UserProfile concept and link to the Address POJO.

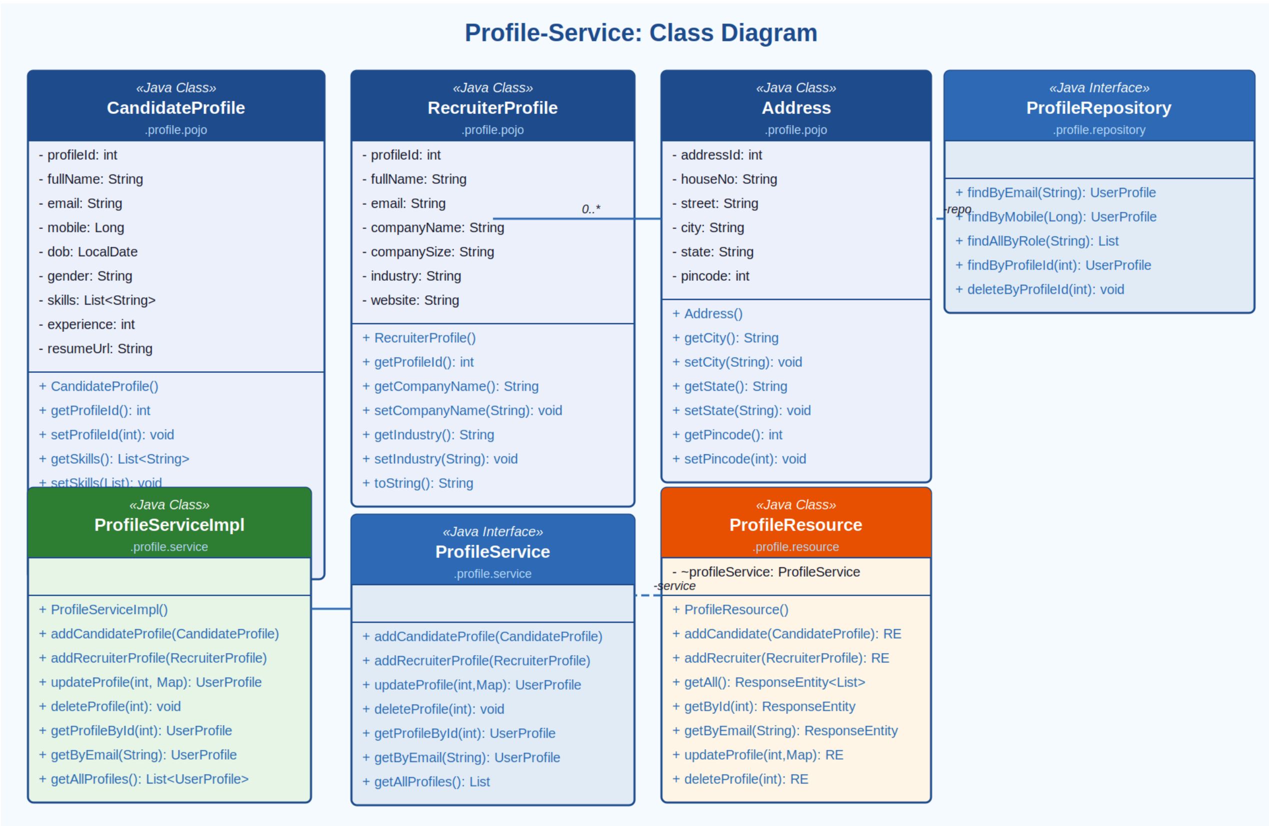

Figure 3: Profile-Service — Class Diagram

Component	Type	Key Attributes / Methods
CandidateProfile	Java Class (POJO)	profileId, fullName, email, mobile, skills: List<String>, experience, resumeUrl, addresses: List<Address>
RecruiterProfile	Java Class (POJO)	profileId, fullName, email, companyName, companySize, industry, website
Address	Java Class (POJO)	houseNo, street, city, state, pincode — shared by both profile types
ProfileRepository	Java Interface	findByEmail(), findByMobile(), findAllByRole(), findByProfileId()
ProfileServiceImpl	Java Class	addCandidateProfile(), addRecruiterProfile(), updateProfile(), deleteProfile()

Component	Type	Key Attributes / Methods
ProfileService	Java Interface	Declares all profile CRUD operations for both user types
ProfileResource	Java Class (REST)	Exposes GET/POST/PUT/DELETE profile endpoints; returns ResponseEntity

4.3 Job-Service

The Job-Service handles the full lifecycle of job postings — creation, retrieval, search, filtering, update, and deletion. Jobs are searchable by title, category, location, salary range, and experience level. Each job tracks its recruiter owner (postedBy) and current status.

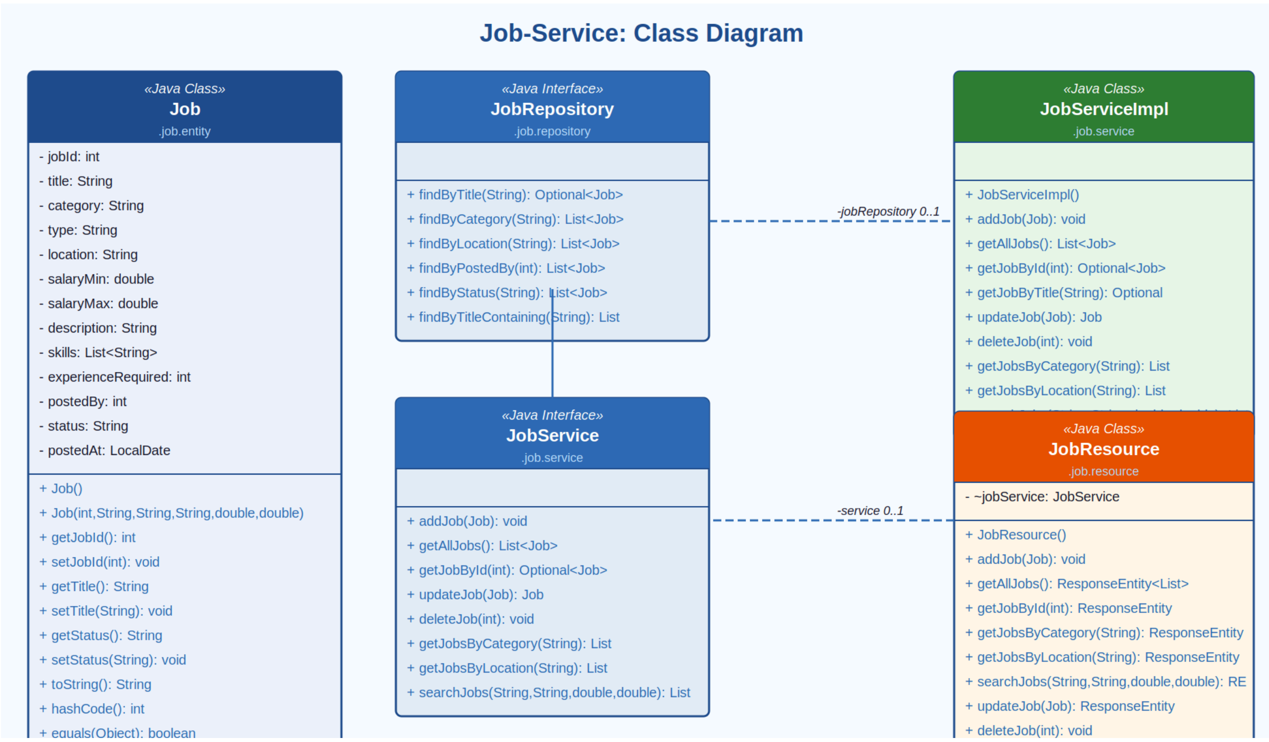

Figure 4: Job-Service — Class Diagram

Component	Type	Key Attributes / Methods
Job	Java Class (Entity)	jobId, title, category, type, location, salaryMin, salaryMax, skills, experienceRequired, postedBy, status, postedAt
JobRepository	Java Interface	findByTitle(), findByCategory(), findByLocation(), findByPostedBy(), findByStatus()
JobServiceImpl	Java Class	addJob(), getAllJobs(), getJobById(), searchJobs(), updateJob(), deleteJob()
JobService	Java Interface	Declares all job CRUD and search/filter operations
JobResource	Java Class (REST)	Exposes /jobs endpoints: GET (list/search/filter), POST, PUT, DELETE

4.4 Application-Service

The Application-Service manages the submission and lifecycle of job applications. It tracks each application's status through the hiring pipeline (Applied → Shortlisted → Interview Scheduled → Offered / Rejected) and supports withdrawal by the candidate.

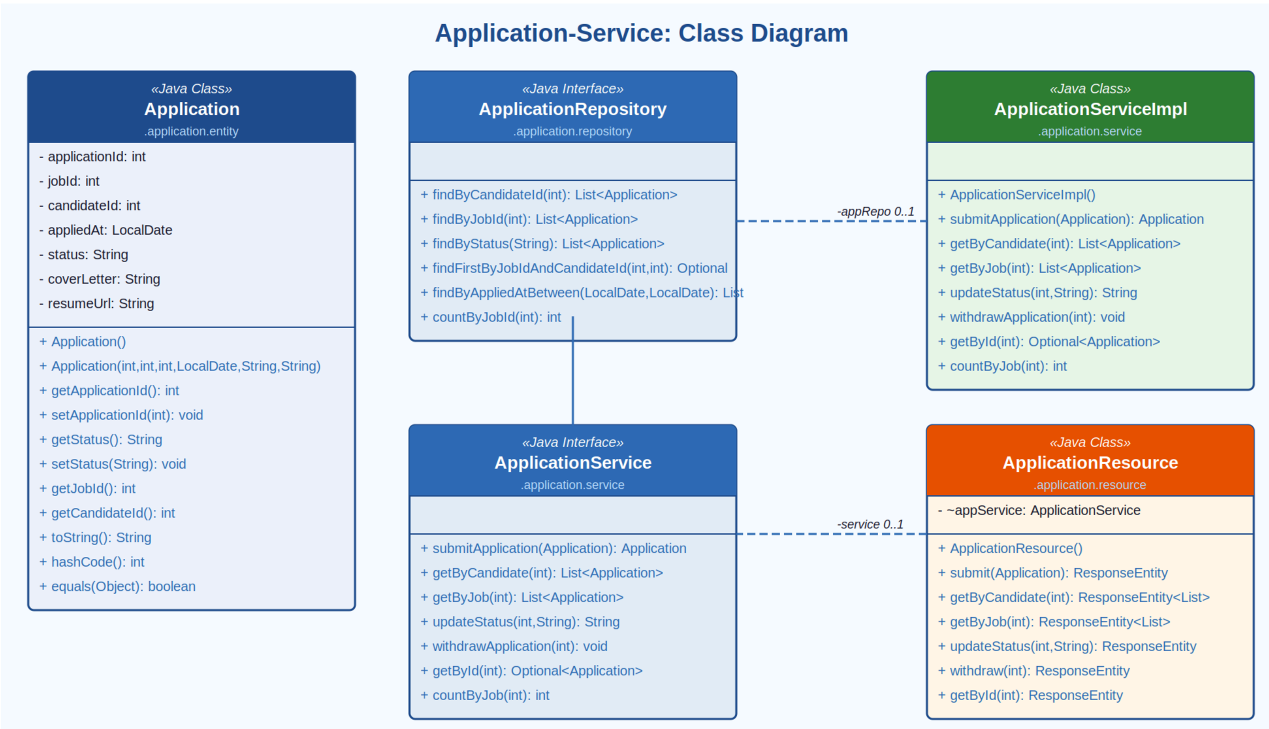

Figure 5: Application-Service — Class Diagram

Component	Type	Key Attributes / Methods
Application	Java Class (Entity)	applicationId, jobId, candidateId, appliedAt, status, coverLetter, resumeUrl
ApplicationRepository	Java Interface	findByCandidateId(), findByJobId(), findByStatus(), findFirstByJobIdAndCandidateId(), countByJobId()
ApplicationServiceImpl	Java Class	submitApplication(), getByCandidate(), getByJob(), updateStatus(), withdrawApplication()
ApplicationService	Java Interface	Declares all application submission, tracking, and management operations
ApplicationResource	Java Class (REST)	Exposes /applications endpoints for submitting, tracking, updating status

4.5 Interview-Service

The Interview-Service handles all interview scheduling logistics. Recruiters can create interview slots with mode (Online / In-Person), meeting link, and notes. Candidates can confirm or request rescheduling. The service maintains interview status through its full lifecycle.

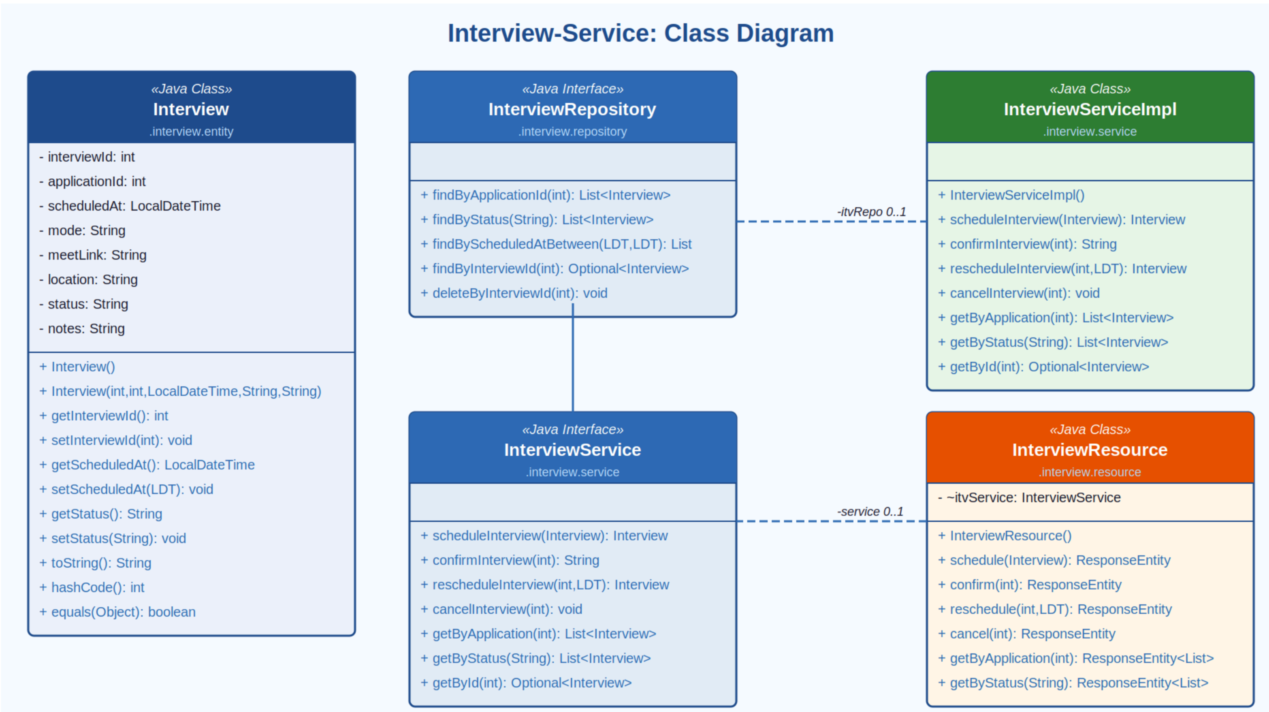

Figure 6: Interview-Service — Class Diagram

Component	Type	Key Attributes / Methods
Interview	Java Class (Entity)	interviewId, applicationId, scheduledAt, mode, meetLink, location, status, notes
InterviewRepository	Java Interface	findByApplicationId(), findByStatus(), findByScheduledAtBetween(), deleteByInterviewId()
InterviewServiceImpl	Java Class	scheduleInterview(), confirmInterview(), rescheduleInterview(), cancelInterview()
InterviewService	Java Interface	Declares all interview scheduling and management operations
InterviewResource	Java Class (REST)	Exposes /interviews endpoints for scheduling, confirming, rescheduling, cancelling

4.6 Notification-Service

The Notification-Service dispatches both in-app and email notifications triggered by system events such as application status changes, interview scheduling, and new job alerts. It tracks read/unread state per user and supports bulk mark-as-read operations.

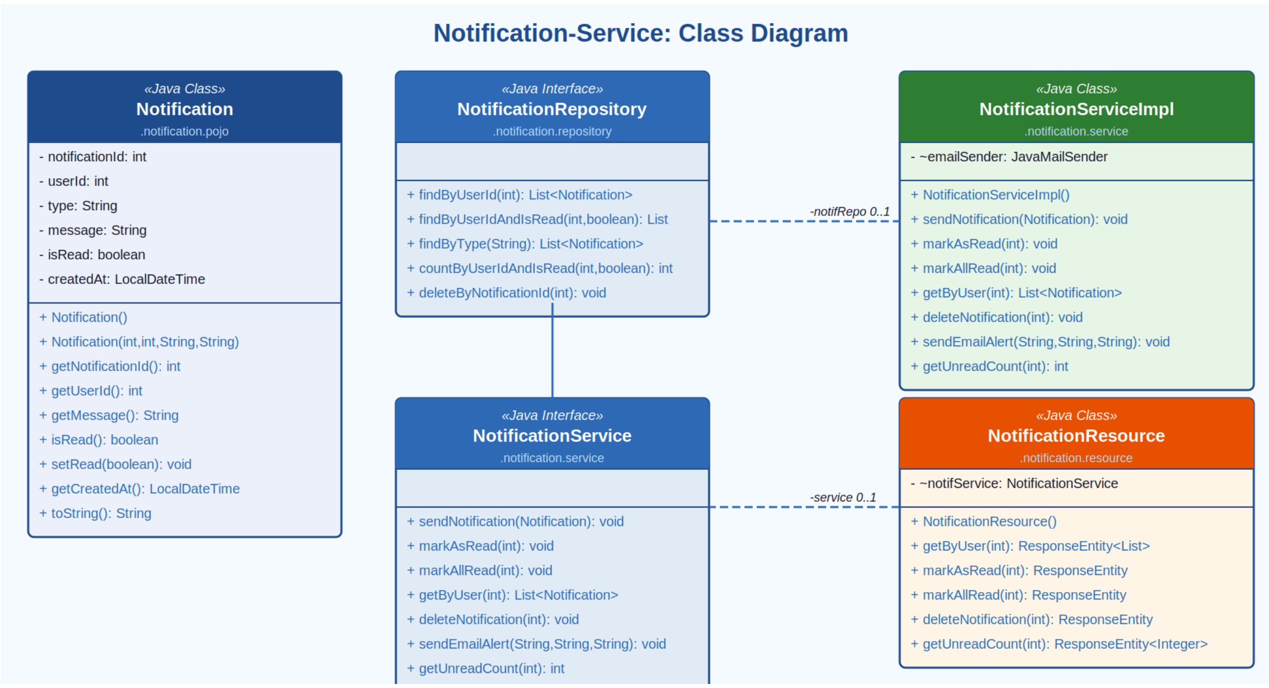

Figure 7: Notification-Service — Class Diagram

Component	Type	Key Attributes / Methods
Notification	Java Class (POJO/Entity)	notificationId, userId, type, message, isRead, createdAt
NotificationRepository	Java Interface	findByUserId(), findByUserIdAndIsRead(), countByUserIdAndIsRead(), deleteByNotificationId()
NotificationServiceImpl	Java Class	sendNotification(), markAsRead(), markAllRead(), getByUser(), sendEmailAlert()
NotificationService	Java Interface	Declares notification dispatch, retrieval, and read-state management
NotificationResource	Java Class (REST)	Exposes /notifications endpoints for listing, marking as read, deleting

4.7 Subscription-Service

The Subscription-Service manages recruiter subscription plans and their associated payment invoices. It supports plan lifecycle operations (subscribe, cancel, renew) and generates invoices on successful payment. Three plan tiers are supported: Free, Professional, and Enterprise.

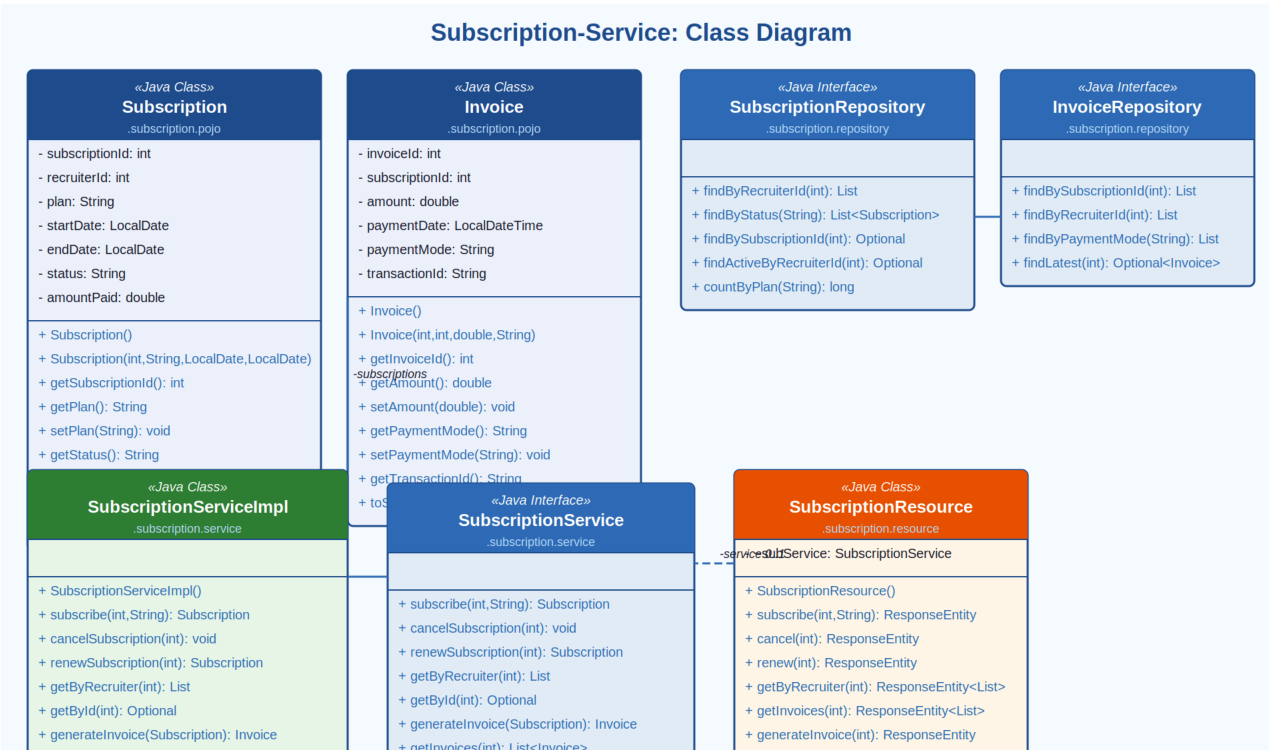

Figure 8: Subscription-Service — Class Diagram

Component	Type	Key Attributes / Methods
Subscription	Java Class (POJO/Entity)	subscriptionId, recruiterId, plan, startDate, endDate, status, amountPaid, isActive()
Invoice	Java Class (POJO/Entity)	invoiceId, subscriptionId, amount, paymentDate, paymentMode, transactionId
SubscriptionRepository	Java Interface	findByRecruiterId(), findByStatus(), findActiveByRecruiterId(), countByPlan()
InvoiceRepository	Java Interface	findBySubscriptionId(), findByRecruiterId(), findLatest()
SubscriptionServiceImpl	Java Class	subscribe(), cancelSubscription(), renewSubscription(), generateInvoice(), getInvoices()
SubscriptionService	Java Interface	Declares subscription lifecycle and invoicing operations

Component	Type	Key Attributes / Methods
SubscriptionResource	Java Class (REST)	Exposes /subscriptions and /invoices REST endpoints

4.8 Analytics-Service

The Analytics-Service provides both recruiter-level and platform-level metrics. Recruiters can view per-job statistics such as view counts, application volumes, view-to-apply ratios, and average time-to-hire. Admins access aggregate platform reports across all users.

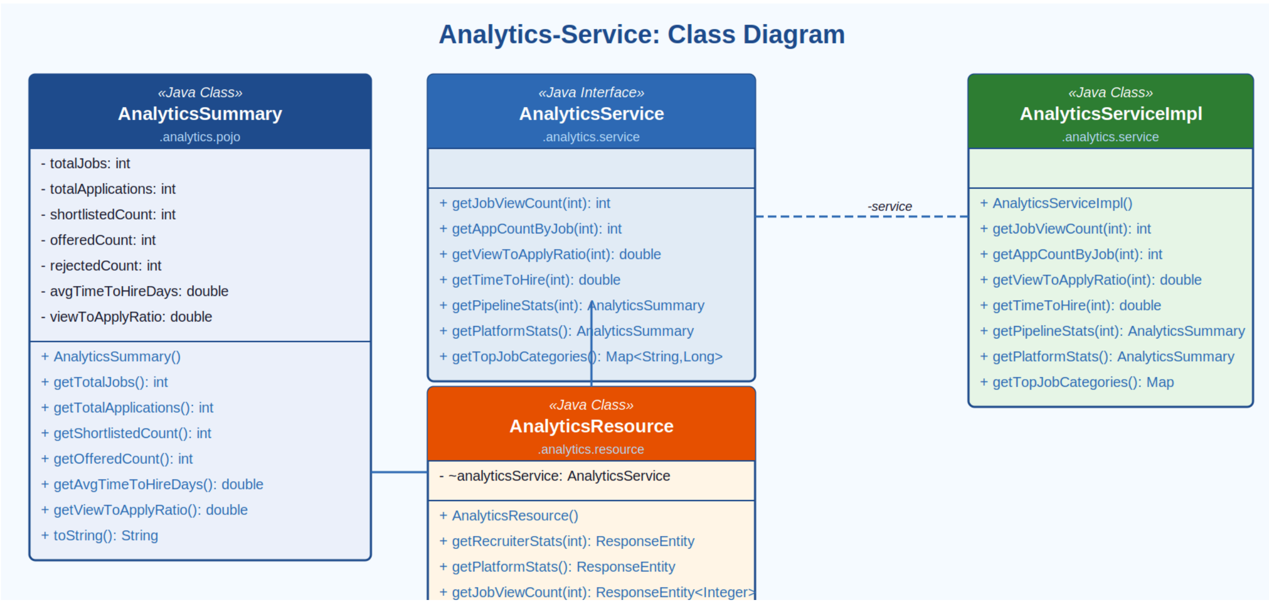

Figure 9: Analytics-Service — Class Diagram

Component	Type	Key Attributes / Methods
AnalyticsSummary	Java Class (POJO)	totalJobs, totalApplications, shortlistedCount, offeredCount, rejectedCount, avgTimeToHireDays, viewToApplyRatio
AnalyticsServiceImpl	Java Class	getJobViewCount(), getAppCountByJob(), getViewToApplyRatio(), getTimeToHire(), getPipelineStats(), getPlatformStats()
AnalyticsService	Java Interface	Declares all analytical query and aggregation operations
AnalyticsResource	Java Class (REST)	Exposes /analytics/recruiter/{id} and /analytics/admin endpoints

4.9 Website-Controller (MVC Layer)

The Website-Controller acts as the MVC web layer integrating all underlying microservices. It contains three controllers — CandidateController, RecruiterController, and AdminController — each wiring together the relevant service dependencies via Spring's @Autowired injection.

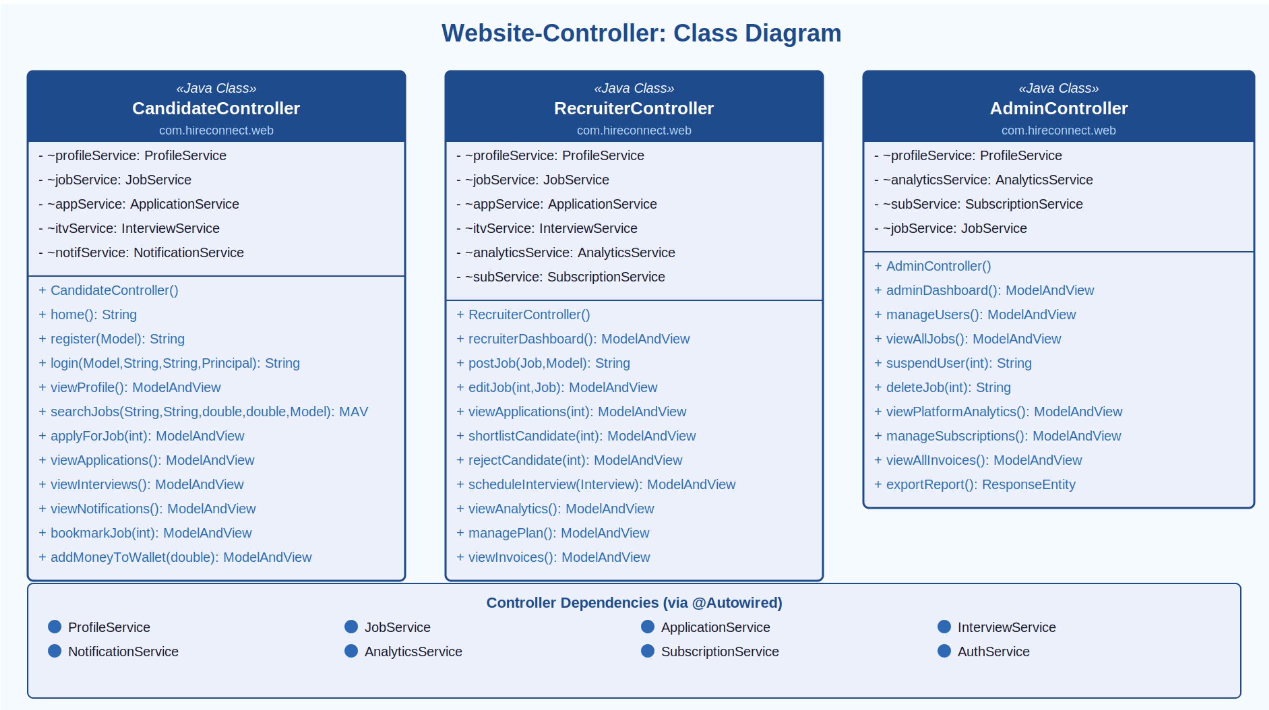

Figure 10: Website-Controller — Class Diagram

Controller	Type	Key Methods
CandidateController	Java Class (MVC)	home(), register(), login(), viewProfile(), searchJobs(), applyForJob(), viewApplications(), viewInterviews(), viewNotifications(), bookmarkJob(), addMoneyToWallet()
RecruiterController	Java Class (MVC)	recruiterDashboard(), postJob(), editJob(), viewApplications(), shortlistCandidate(), rejectCandidate(), scheduleInterview(), viewAnalytics(), managePlan(), viewInvoices()
AdminController	Java Class (MVC)	adminDashboard(), manageUsers(), viewAllJobs(), suspendUser(), viewPlatformAnalytics(), manageSubscriptions(), viewAllInvoices(), exportReport()

5. Microservices Architecture Overview

HireConnect is designed as a set of independently deployable microservices, each responsible for a clearly defined domain. Inter-service communication uses REST (via RestTemplate) for synchronous calls and RabbitMQ for asynchronous notification events.

Microservice	Base Package	Primary Domain
auth-service	com.hireconnect.auth	Authentication, JWT, OAuth2 (GitHub)
profile-service	com.hireconnect.profile	Candidate & Recruiter profile management
job-service	com.hireconnect.job	Job postings, search, and filtering
application-service	com.hireconnect.application	Job applications and status tracking
interview-service	com.hireconnect.interview	Interview scheduling and management
notification-service	com.hireconnect.notification	In-app and email notifications
subscription-service	com.hireconnect.subscription	Recruiter plans and payment invoices
analytics-service	com.hireconnect.analytics	Hiring metrics and platform reporting
hireconnect-web	com.hireconnect.web	MVC Web Controller (frontend integration)

6. Non-Functional Requirements

Category	Requirement
Performance	Job search results must return within 2 seconds for up to 10,000 concurrent users
Scalability	Each microservice must be independently scalable using Docker/Kubernetes orchestration
Security	All passwords stored as bcrypt hashes; JWT tokens expire in 24 hours; OAuth2 for GitHub login
Availability	99.9% uptime SLA; health-check endpoints exposed for each service
Data Integrity	Database transactions must be ACID-compliant; application status changes are atomic
Usability	Responsive UI supporting desktop and mobile browsers; WCAG 2.1 AA accessibility compliance
Maintainability	Each service independently deployable; RESTful API contracts versioned under /api/v1/

7. Proposed Technology Stack

Layer	Technology
Backend Framework	Java Spring Boot (Spring MVC, Spring Security, Spring Data JPA)
Authentication	JWT (JSON Web Tokens), Spring Security OAuth2, GitHub OAuth
Database	MySQL (primary relational store); Redis (session and notification caching)
Search	Elasticsearch (job search and full-text filtering)
Frontend	Thymeleaf templates (MVC) or React.js (SPA) for candidate/recruiter dashboards
File Storage	AWS S3 or local file server for resume PDF storage
Messaging	RabbitMQ for asynchronous notification dispatch
Containerisation	Docker with Docker Compose; Kubernetes for production orchestration
API Documentation	Swagger / OpenAPI 3.0
CI/CD	GitHub Actions or Jenkins pipeline for automated build and deployment

8. Glossary

Term	Definition
Candidate	A registered job seeker who applies for job openings on HireConnect
Recruiter	An employer or HR professional who posts jobs and manages applications
ATS	Applicant Tracking System — tool used to manage candidates through the hiring pipeline
JWT	JSON Web Token — a compact means of representing claims for authentication
OAuth	Open Authorization — open standard for access delegation, used here for GitHub login
Pipeline	The sequence of stages a candidate moves through during the hiring process
Shortlisting	Recruiter action to mark a candidate as qualified for further consideration
Resume Parsing	Automated extraction of structured data (skills, experience) from a PDF resume
Microservice	An independently deployable service responsible for a single business domain
POJO	Plain Old Java Object — a simple Java class without framework dependencies